

MAX PLANCK INSTITUTE
FOR GRAVITATIONAL PHYSICS
(Albert Einstein Institute)



When may Pulsar Timing Arrays observe signatures of individual supermassive black hole binaries?

Uncovering the origin of the gravitational wave background with new pulsar timing array experiments

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Introduction

Methodology

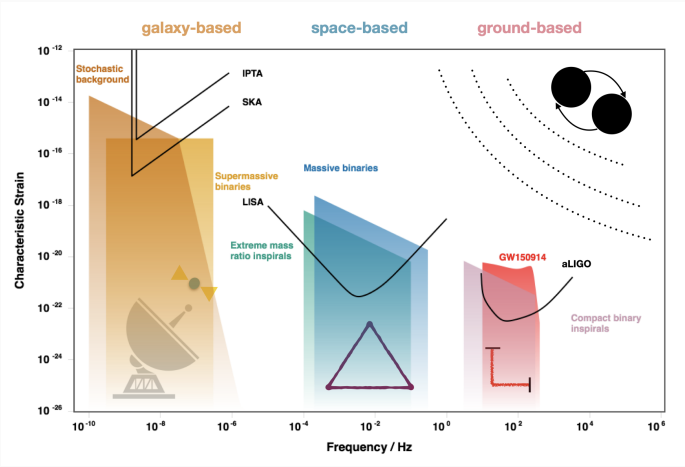
Simulations

Results

*extra



The gravitational-wave spectrum



Credit: Mingarelli and Mingarelli, GWPlotter.

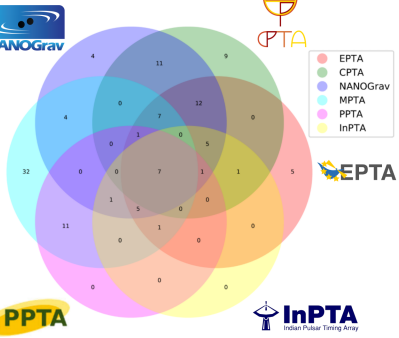
The International Pulsar Timing Array Collaboration



NANOGrav

EPTA

SKAO
MEERTIME



EPTA

PPTA

InPTA
Indian Pulsar Timing Array

Credit: Kuo Liu

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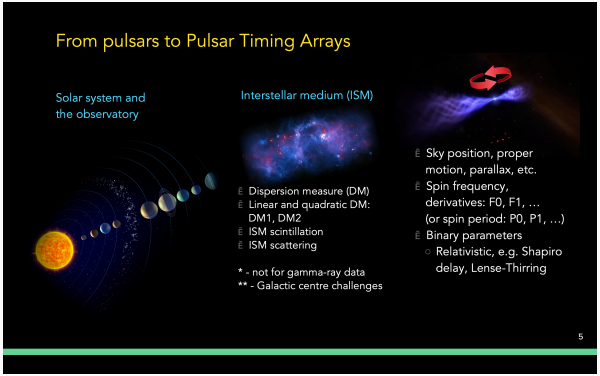
From one pulsar to an array

Millisecond pulsars are stable clocks. We measure times of arrival (ToAs) $\mathbf{t}_a$ and subtract a deterministic *timing model* $\mathbf{t}_p$:

$$\boldsymbol{\delta t} = \mathbf{t}_a - \mathbf{t}_p, \qquad \mathbf{t}_p = \mathbf{M}\boldsymbol{\varepsilon} + \mathbf{F}\mathbf{a},$$

where $\boldsymbol{\delta t}$ is the *timing residual*.

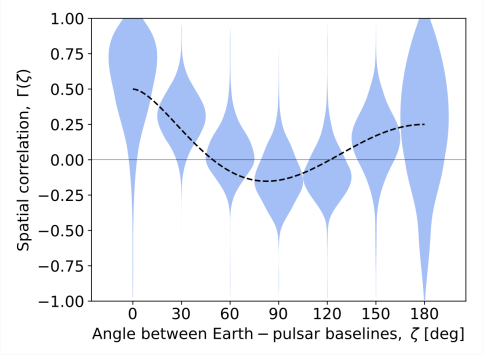
Noise	Form
White	$\mathrm{EFAC}^2\sigma_a^2 + \mathrm{EQUAD}^2$
DM var.	$\propto \nu^{-2}$
Spin (red)	$P_a(f) \propto f^{-\gamma_a}$



The common signal: Hellings–Downs correlations

For an isotropic, unpolarised background the expected inter-pulsar correlation depends only on the angular separation ζ_{ab}:

$$\Gamma_{\mathrm{HD}}(\zeta) = \frac{1}{2} - \frac{x}{4} + \frac{3x}{2}\ln x + \frac{1}{2}\delta_{ab}, \quad x = \frac{1-\cos\zeta}{2}.$$

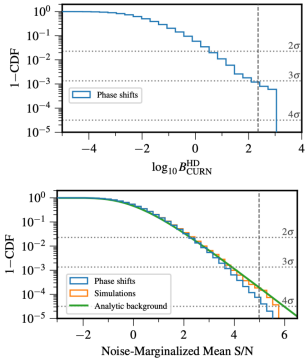


Goncharov, Sardana, Sesana, Tomson, et al. (EPTA), *Nature Communications*, arXiv:2409.03627.

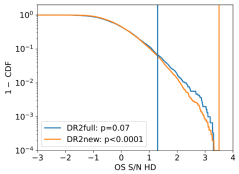
Background evaluation for PTA results

NANOGrav

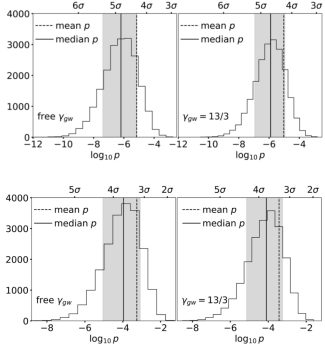
arXiv: 2306.16213



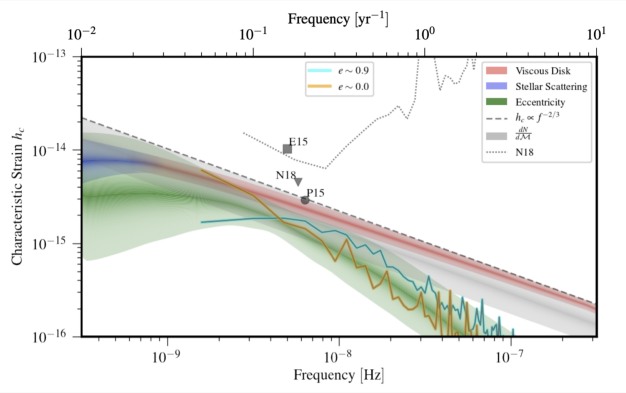
EPTA arXiv: 2306.16214



Yu and Allen: 2512.08666
Data from 5 PTAs



Where does the background come from?



Sarah Burke-Spolaor, Stephen R. Taylor, Maria Charisi, et al. *The astrophysics of nanohertz gravitational waves*, A&A Rev. **27**, 5 (2019).

Astrophysical: a population of inspiralling SMBHBs. For circular, GW-driven binaries

$$h_c(f) \propto f^{-2/3} \quad \Longleftrightarrow \quad \gamma = 13/3.$$

Near $f \sim \mathrm{yr}^{-1}$ the power is dominated by a handful of the brightest binaries.

Cosmological: phase transitions, cosmic strings, inflationary relics.

Both can be tuned to the same power law, so the amplitude and spectral index *alone* cannot decide between them.



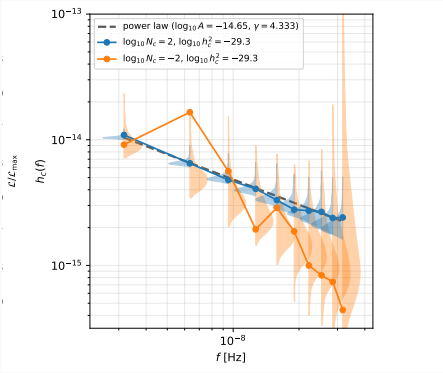
The discriminant: a finite population is non-Gaussian

In each Fourier bin the background is the incoherent sum of N binaries,

$$h_t^2 = \sum_{i=1}^{N} h_{s,i}^2, \qquad N \sim \text{Poisson}.$$

- Many sources per bin: central limit theorem, hence a smooth Gaussian background on a power law.
- Few sources per bin: heavy-tailed bin-to-bin fluctuations, with spikes where a bright binary lands.

Cosmological backgrounds have no such granularity, so non-Gaussianity is a direct test of the SMBHB origin.



Two draws with the same mean spectrum: $\log_{10} N_c = 2$ (blue, near-Gaussian) and $\log_{10} N_c = -2$ (orange, strongly discrete).

From first principles: toward a definitive GWB model

Deliverables

- Discovery of the origin of the GWB
- Global fit to all GW signals
- Population inference on-the-fly

What is available

- *Spectrum:* power-law, free spectrum, t-process
- *Signals:* anisotropy, CWs, HD, holedeck

What is required

- Definitive models
- First principles

Our approach

Start from the SMBHB Poisson process, derive an astrophysically informed *prior* on the free spectrum, and infer hyperparameters with a hierarchical Bayesian framework:

$$p(\boldsymbol{\rho} \mid \boldsymbol{\Lambda}) = \prod_k p(\rho_k \mid N_c, h_c^2), \qquad \boldsymbol{\Lambda} = (N_c, h_c^2).$$

- Only two hyperparameters, both with clear physical meaning.
- The Gaussian background is recovered as $N_c \to \infty$.
- Non-Gaussianity enters through the population, not ad hoc mixture likelihoods on the Fourier coefficients.



Where we stand

Observational status

- NANOGrav compared the 15 yr free spectrum with semi-analytic SMBHB realisations: two excursions from the power law, at $2\,\mathrm{nHz}$ and $16\,\mathrm{nHz}$, both at roughly 2σ.
- No significant departure from Gaussianity has been reported.
- Continuous-wave searches: candidates, but no detection.

What nobody has quoted

How often should a detection of discreteness be expected in future data sets?

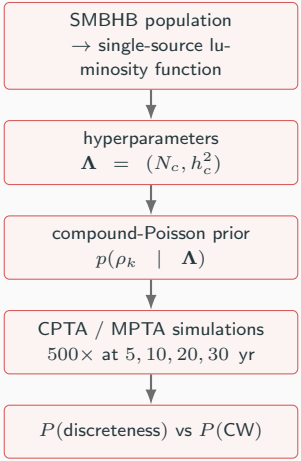


This talk

Question. When will PTAs see the signature of *individual* SMBHBs in the nanohertz background?

- Discreteness of the unresolved background: Savage–Dickey Bayes Factor $\ln \mathcal{B} > 2$ against the Gaussian limit $\log_{10} N_c = 3$.
- Continuous waves from the single loudest binary: $\mathrm{SNR} > 2$.

Method. Realistic CPTA and MPTA simulations with published noise budgets, observation spans of 5, 10, 20, $30\,\mathrm{yr}$, 500 independent realisations each, analysed with a hierarchical Bayesian model.



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PTA likelihood

Post-fit residuals $\boldsymbol{\delta t} = \mathbf{t}_a - \mathbf{t}_p$ with $\mathbf{t}_p = \mathbf{M}\boldsymbol{\varepsilon} + \mathbf{F}\mathbf{a}$. Marginalising analytically over $\boldsymbol{\varepsilon}$ and $\mathbf{a}$,

$$\mathcal{L}(\boldsymbol{\delta t} \mid \boldsymbol{\theta}) = \frac{\exp\!\left(-\frac{1}{2}\boldsymbol{\delta t}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{\delta t}\right)}{\sqrt{2\pi \det \mathbf{C}}}.$$

A $T_{\rm obs} > 10\,\mathrm{yr}$ pulsar has $\gtrsim 10^5$ ToAs, so $\mathbf{C}$ is handled through the Woodbury identity in low-rank form,

$$\mathbf{C} = \mathbf{N} + \mathbf{T}\mathbf{B}\mathbf{T}^{\mathsf{T}}, \qquad \mathbf{T} = (\mathbf{M}, \mathbf{F}),$$

$$\mathbf{B} = \begin{pmatrix} B_{\rm TM} & 0 \\ 0 & \Gamma_{ab} \otimes \boldsymbol{\Phi} \end{pmatrix}.$$

- $\mathbf{N}$: white-noise (diagonal) covariance.
- $B_{\rm TM} \to \infty$, $B_{\rm TM}^{-1} = 0$, uniform prior for the timing model.
- $\Gamma_{ab} = \Gamma_{\rm HD}$ for the GWB, $\Gamma_{ab} = \mathbb{I}$ for CURN.
- $\boldsymbol{\Phi}$: single-pulsar Fourier-domain covariance, i.e. the power spectrum $P_r(f_i)$.



The free spectrum model

Fourier power in bin i, linked to the characteristic strain:

$$P_r(f_i)\,\Delta f = \rho_i^2, \qquad P_r(f_i) = \frac{h_c^2(f_i)}{12\pi^2 f_i^3}, \qquad \Delta f = \frac{1}{T_{\rm obs}}.$$

Each ρ_k sets the prior on the cosine and sine Fourier coefficients in bin k,

$$p(a_k, b_k \mid \rho_k) = \mathcal{N}(a_k; 0, \rho_k^2)\,\mathcal{N}(b_k; 0, \rho_k^2), \qquad \mathbf{\Phi}_k = \rho_k^2\,\mathbf{I}_2.$$

Marginalising over $\mathbf{a}$, $\mathbf{C}(\boldsymbol{\rho})$ and $\mathcal{L}(\boldsymbol{\delta t} \mid \boldsymbol{\theta}, \boldsymbol{\rho})$ depends on $\boldsymbol{\rho}$ only through $\mathbf{\Phi}$. The free-spectrum model assigns each bin an independent prior on $\log_{10}\rho_k$,

$$p(\log_{10}\boldsymbol{\rho}) = \prod_{k=1}^{N_f} \mathrm{Uniform}(\rho_k^{\min}, \rho_k^{\max}), \qquad p(\boldsymbol{\theta}, \boldsymbol{\rho} \mid \boldsymbol{\delta t}) \propto \mathcal{L}(\boldsymbol{\delta t} \mid \boldsymbol{\theta}, \boldsymbol{\rho})\,p(\boldsymbol{\theta})\,p(\boldsymbol{\rho}).$$



Hierarchical inference

Keep the likelihood, replace the prior:

$$p(\boldsymbol{\theta}, \boldsymbol{\rho} \mid \boldsymbol{\delta t}) \propto \mathcal{L}(\boldsymbol{\delta t} \mid \boldsymbol{\theta}, \boldsymbol{\rho})\, p(\boldsymbol{\theta})\, p(\boldsymbol{\rho})$$

$$\Downarrow \qquad p(\boldsymbol{\rho}) \longrightarrow p(\boldsymbol{\rho} \mid \boldsymbol{\Lambda})\, p(\boldsymbol{\Lambda})$$

$$p(\boldsymbol{\theta}, \boldsymbol{\rho}, \boldsymbol{\Lambda} \mid \boldsymbol{\delta t}) \propto \mathcal{L}(\boldsymbol{\delta t} \mid \boldsymbol{\theta}, \boldsymbol{\rho})\, p(\boldsymbol{\theta})\, p(\boldsymbol{\rho} \mid \boldsymbol{\Lambda})\, p(\boldsymbol{\Lambda})$$

- $\boldsymbol{\Lambda} \equiv (N_c, h_c^2)$: two hyperparameters describing the SMBHB population, with hyperprior $p(\boldsymbol{\Lambda})$.

Since ρ_k is just a rescaling of the total squared strain h_t^2 in bin k, a population model that predicts $p(h_t^2 \mid \boldsymbol{\Lambda})$ gives the conditional prior directly:

$$p(\log_{10} \rho_k \mid \boldsymbol{\Lambda}) = p(h_t^2 \mid \boldsymbol{\Lambda}) \left| \frac{dh_t^2}{d \log_{10} \rho_k} \right| = p(h_t^2 \mid \boldsymbol{\Lambda}) \times 24\pi^2 f_k^3 T_{\rm obs}\, \rho_k^2 \ln 10.$$



Step 1: the single-source luminosity function

$$\frac{dN}{d\log h_s^2} = \int dz \int dq \underbrace{\left|\frac{\partial \log M}{\partial \log h_s^2}\right|}_{=3/10} \underbrace{\frac{dn}{d\log M}}_{\text{mass function}} \underbrace{p_z(z)\, p_q(q)}_{\text{merger demographics}} \underbrace{\frac{dV_c}{d\log f}}_{\propto M_c^{-5/3} f^{-8/3}} \underbrace{\frac{\Delta f}{f}}_{\text{bin width}} .$$

- $p(h_s^2) \propto dN/dh_s^2(h_s^2; f, \Delta f)$.
- $p_z(z) \propto z^B e^{-z/Z_s}$, $p_q(q) \propto q^C$.
- $dV_c/d\log f$ from the GW-driven residence time ($M_c = M\eta^{3/5}$, $\eta = q/(1+q)^2$).
- The Jacobian $3/10$ from the inclination- and polarisation-averaged strain of one binary.
- $dn/d\log M$ on the quadrupole map $\tilde{h}_s^2(M, q, z, f) = h_s^2$.



Step 1: the single-source luminosity function

The mass function follows from the local velocity-dispersion function (VDF),

$$\frac{dn}{d\log\sigma} = \phi_\sigma \left(\frac{\sigma}{\sigma_{\rm ref}}\right)^{a_\sigma} \exp\left[-\left(\frac{\sigma}{\sigma_{\rm ref}}\right)^{b_\sigma}\right],$$

mapped to M via the M–σ relation plus Gaussian scatter,

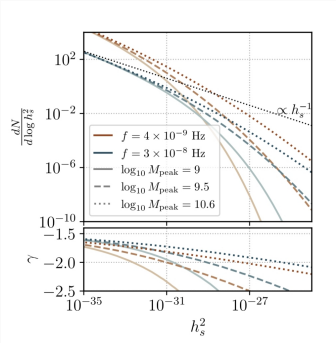
$$\log_{10} M \;\mid\; \sigma \sim \mathcal{N}\Big(a_{M\sigma} + b_{M\sigma}\log_{10}\tfrac{\sigma}{200\,\mathrm{km/s}},\ \epsilon^2_{M\sigma}\Big).$$

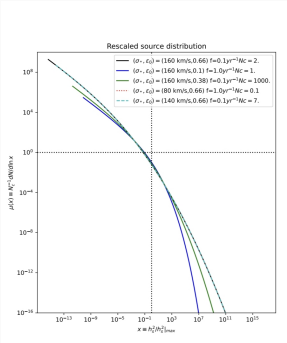
Convolving the two gives $dn/d\log M$.

Quantity	Role	Value
B	p_z index	0.5
Z_s	p_z cutoff	0.33
C	p_q index	-1
ϕ_σ	VDF norm.	2.611×10^{-2}
$\sigma_{\rm ref}$	VDF scale	$170\,\mathrm{km\,s^{-1}}$
(a_σ, b_σ)	VDF shape	$(0.41,\ 2.59)$
$(a_{M\sigma}, b_{M\sigma})$	M–σ	$(8.32,\ 5.64)$
$\epsilon_{M\sigma}$	M–σ scatter	0.38



Step 1: the single-source luminosity function





G. Sato-Polito and M. Zaldarriaga, Phys. Rev. D **111**, 023043 (2025), arXiv:2406.17010.

Step 2: two scales and one fixed shape

Factor dN/dh_s^2 into overall scales plus a dimensionless shape:

$$h^2_{\rm peak} = \arg\max_{h_s^2}\left[h_s^4\, dN/dh_s^2\right], \qquad N_c = \frac{\displaystyle\int h_s^2\,\frac{dN}{dh_s^2}\,dh_s^2}{h^2_{\rm peak}}, \qquad h_c^2 = N_c\, h^2_{\rm peak}.$$

Measuring strain in units of the peak,

$$x \equiv \frac{h_s^2}{h^2_{\rm peak}}, \qquad \mu(x) = \frac{h_s^2\,(dN/dh_s^2)}{N_c} \qquad \Longrightarrow \qquad \frac{dN}{dx} = \frac{N_c\,\mu(x)}{x}.$$



Step 3: compound-Poisson superposition

In one frequency bin, $h_t^2 = \sum_{i=1}^{N} h_{s,i}^2$, and we work with $S \equiv h_t^2/h_{\rm peak}^2$. With $\frac{dN}{dx} = \frac{N_c\,\mu(x)}{x}$, split the population at $S_{\rm min}$ so that exactly one source is expected above it:

$$\int_{S_{\rm min}}^{\infty} \frac{dN}{dx}\,dx = 1.$$

Faint ($x < S_{\rm min}$): many sources, replaced by a Gaussian via the CLT,

$$\bar{S}_{\rm f} = \int_{x<S_{\rm min}} \frac{dN}{dx} x\,dx, \quad \sigma_{\rm f}^2 = \int_{x<S_{\rm min}} \frac{dN}{dx} x^2\,dx.$$

Bright ($x > S_{\rm min}$): $\mathcal{O}(1)$ sources, keep the exact compound-Poisson characteristic function,

$$\varphi_{\rm b}(\omega) = \exp\Big[\int_{S_{\rm min}}^{\infty} \big(e^{i\omega x} - 1\big) \frac{dN}{dx}\,dx\Big].$$

Characteristic functions multiply, so the total PDF is a single inverse FFT:

$$p(S) = \mathcal{F}^{-1}\Big[\exp\big(\mathcal{F}[dN/dS] - \mathcal{F}[dN/dS]\big|_{\omega=0} - \tfrac{1}{2}\sigma_{\rm f}^2\omega^2 - i\bar{S}_{\rm f}\omega\big)\Big].$$

Then $h_t^2 = S\,h_{\rm peak}^2$ gives $p(h_t^2 \mid N_c, h_c^2)$, and hence $p(\rho_k \mid \mathbf{\Lambda})$.



Discreteness grows towards higher frequencies

The hyperparameters are defined at $f_{\rm ref} = 1\,\mathrm{yr}^{-1}$ and must be propagated across the band. For a circular, GW-driven population ($\Omega_{\rm gw} \propto f^{2/3}$):

$$h_c^2(f) = \frac{3H_0^2}{2\pi^2}\frac{\Omega_{\rm gw}(f)}{f^2} = h_{c,\rm ref}^2\left(\frac{f}{f_{\rm ref}}\right)^{-4/3},$$

while the angle-averaged contribution of a single binary in a bin of width $\Delta f = 1/T_{\rm obs}$,

$$h_{\rm peak}^2 = \frac{32}{5}\Big(\frac{GM_c}{c^2}\Big)^{10/3}\eta^2\frac{(1+z)^{4/3}}{\chi^2(z)}\Big(\frac{\pi f}{c}\Big)^{4/3}\frac{f}{\Delta f},$$

scales as $f^{4/3} \times f = f^{7/3}$.

Therefore

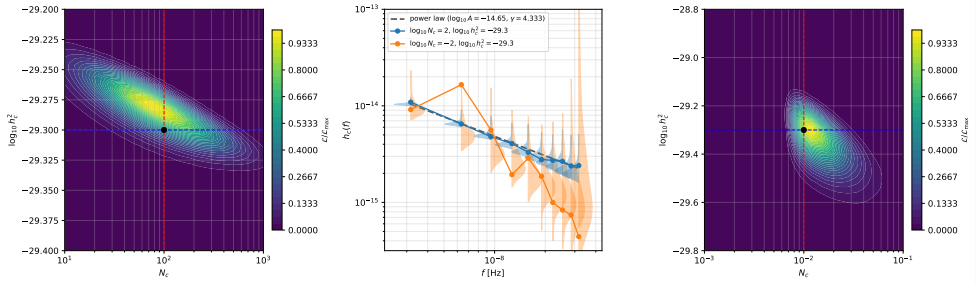
$$N_c(f) = \frac{h_c^2(f)}{h_{\rm peak}^2(f)} = N_{c,\rm ref}\left(\frac{f}{f_{\rm ref}}\right)^{-11/3},$$

with $-11/3 = -4/3 - 7/3$.

- Low-frequency bins contain many effective sources: smooth, Gaussian, and this is where h_c^2 is measured.
- High-frequency bins are built from a few binaries: this is where discreteness lives — and where the white-noise floor fights back.

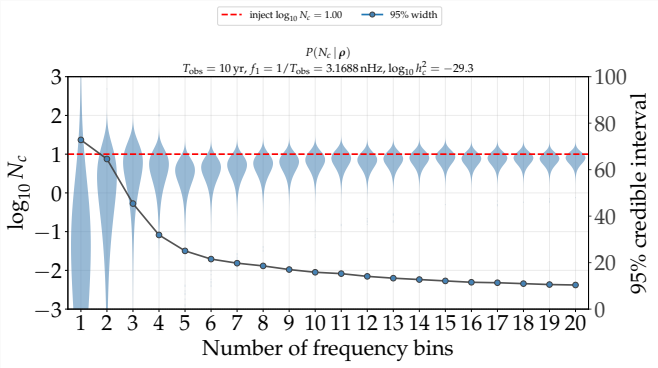


How much can we hope to learn? (noise-free limit)



Using $p(\rho_k \mid \mathbf{\Lambda})$ evaluated at the injected $\{\rho_k\}$ as the likelihood isolates the information carried by the astrophysical model alone—no PTA noise. Same $\log_{10} h_c^2 = -29.3$, but $\log_{10} N_c = 2$ (left, near-Gaussian) and $\log_{10} N_c = -2$ (right, strongly discrete). The low-N_c realisation gives a much tighter constraint on N_c; the near-Gaussian one barely constrains it at all.

How many frequency bins do we need?



Noise-free test at $T_{\rm obs} = 10\,\mathrm{yr}$ with injected $\log_{10} N_c = 1$ and $\log_{10} h_c^2 = -29.3$.

- The posterior on $\log_{10} N_c$ tightens about the injection as bins are added.
- The 95% credible width drops steeply over the first ~ 5 bins, then flattens.
- $\Rightarrow$ use $\gtrsim 10$ frequencies; in practice we analyse 30.

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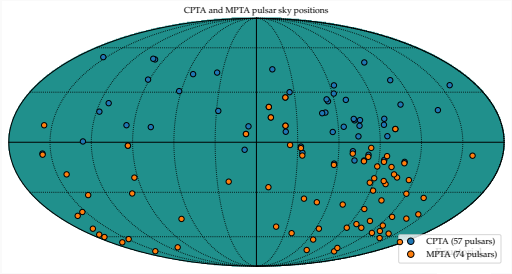
Mock data: what goes in

Pipeline built on ENTERPRISE (signal model), PINT (fast timing-model evaluation) and DISCOVERY (JAX/GPU Bayesian inference).

(a) Prior-draw validation. Draw (N_c, h_c^2) from the same hyperprior used at inference, then $\{\rho_k\}$ from $p(\rho_k \mid \mathbf{\Lambda})$. This tests that the hierarchical pipeline recovers what it injects.

(b) Fixed-hyperparameter forecast. Fix (N_c, h_c^2) at astrophysically conservative values ($N_c = 10$ at yr^{-1}) and generate many independent spectrum and noise realisations. This isolates the sampling variance of the discrete background and yields the detection probabilities.

	CPTA DR1	MPTA DR1
Pulsars	57	74
Realisations	500	500
T_{obs}	5, 10, 20, 30 yr	
N_f	30	



Mock data: the noise budget

	CPTA DR1	MPTA DR1
White noise		
σ_a per pulsar	post-fit residual RMS	subband whitened RMS (else subband full RMS)
EFAC	fixed at 1.0	fixed at 1.0
EQUAD / ECORR	not used	not used
Intrinsic red (spin) noise, $P_a(f) = A_a^2/(12\pi^2)\,(f/f_{\rm yr})^{-\gamma_a}$, $N_{\rm sn} = 30$ *Fourier modes*		
Detected	11 pulsars: DR1 $(\log_{10} A_{\rm RN,DMGP}, \gamma)$	4 pulsars: DR1 $(\log_{10} A_{\rm red}, \gamma)$
Not detected	$\log_{10} A_a \sim \mathrm{U}[-16,\, \log_{10} A_{\rm UL}]$	$\log_{10} A_a \sim \mathrm{U}[-16,\, -12.71]$
Missing γ	$\gamma_a \sim \mathrm{U}[0.32,\, 3.97]$	$\gamma_a \sim \mathrm{U}[0.32,\, 3.97]$
Recovery priors		
Spin noise	$\log_{10} A_a \sim \mathrm{U}[-16, -11]$, $\gamma_a \sim \mathrm{U}[0.32, 3.97]$	
Hyperparameters	$\log_{10} N_c \sim \mathrm{U}[-3, 3]$, $\log_{10} h^2_{\rm peak} \sim \mathrm{U}[-34, -27]$	

The recovery prior on $\log_{10} A_a$ is deliberately wider than the injection range, so that upper-limit draws above -12.71 stay inside the analysis support.



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The Savage–Dickey Bayes Factor

We take N_c as the detection statistic and identify the Gaussian hypothesis with the prior upper bound $\log_{10} N_c = 3$: a population this dense is indistinguishable from a Gaussian GWB. The evidence for discreteness is the Savage–Dickey density ratio at that bound,

$$\mathcal{B} = \frac{\pi(\log_{10} N_c = 3)}{\mathcal{P}(\log_{10} N_c = 3 \mid d)}.$$

- Evaluated by histogramming the $\log_{10} N_c$ samples over the prior range: $\mathcal{B}$ is the average bin count divided by the count in the bin abutting the bound.
- Realisations with an *empty* last bin saturate at $\ln \mathcal{B} > \ln N_{\rm avg} = 5.12$.
- Detection probability = fraction of realisations with $\ln \mathcal{B} > 2$.

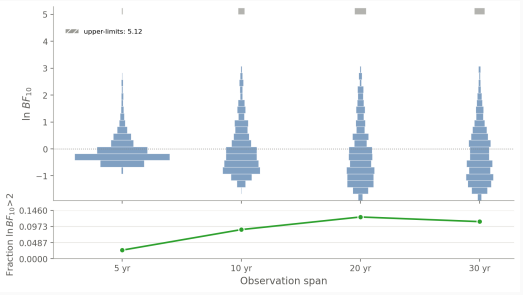
Why $\ln \mathcal{B} > 2$ and $\mathrm{SNR} > 2$ are comparable thresholds: for a signal linear in its amplitude in Gaussian noise, $2 \ln \mathcal{L}_{\max} = \mathrm{SNR}^2$, and in the Laplace approximation

$$\ln \mathcal{B} \simeq \frac{\mathrm{SNR}^2}{2} - \ln \mathcal{O}, \qquad \mathrm{SNR} \simeq \sqrt{2 \ln \mathcal{B}},$$

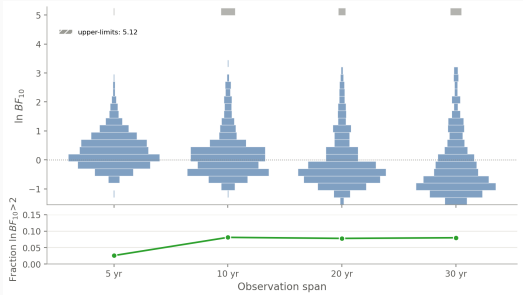
with $\mathcal{O} \geq 1$ the Occam factor. Both cuts correspond to a $\sim 2\sigma$ departure from the null hypothesis.



Result 1: evidence for discreteness vs observing span



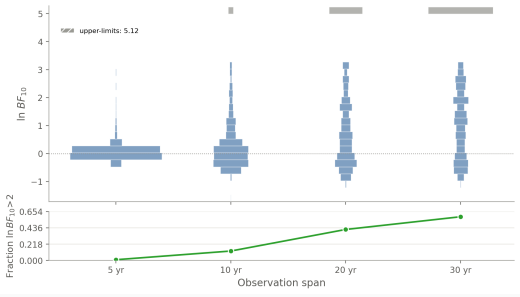
CPTA — 57 pulsars, 500 realisations



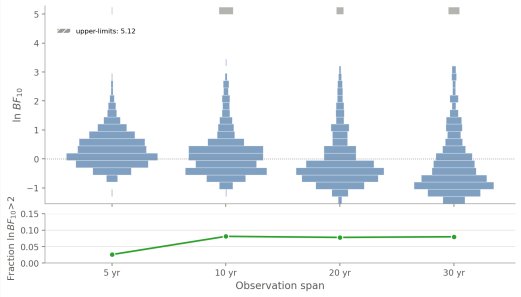
MPTA — 74 pulsars, 500 realisations

Blue: the distribution of $\ln \mathcal{B}$ over realisations at each span. Orange: realisations at the numerical ceiling, with no posterior support at the Gaussian bound. Curves (right axis): the fraction exceeding $\ln \mathcal{B} > 1$ and > 2. Longer baselines open more low-frequency bins, so bright-source fluctuations become distinguishable more often and the high-$\mathcal{B}$ tail grows.

MPTA: effect of the injected N_c



$\log_{10} N_c = -1$



$\log_{10} N_c = 1$

MPTA: group posterior across realisations

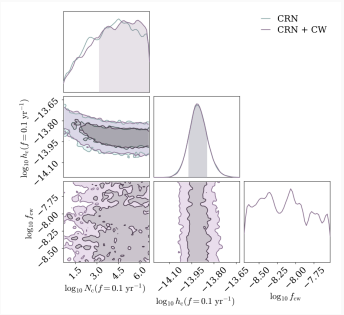


Result 2: discreteness vs continuous waves

Detection probabilities for injected $N_c = 10$ at yr^{-1}.

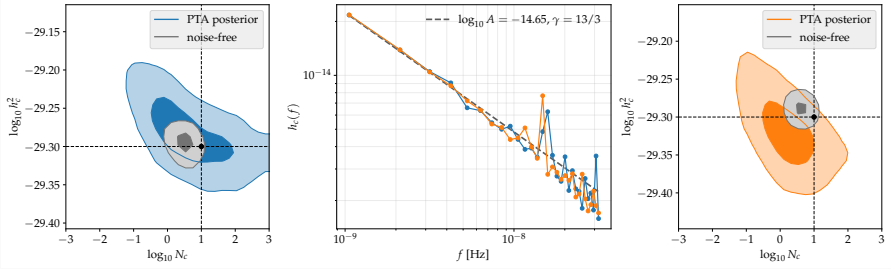
Prob. [%]	5 yr	10 yr	20 yr	30 yr
CW (CPTA)	$4.1^{+1.1}_{-1.0}$	$15.8^{+2.2}_{-1.9}$	$45.2^{+3.0}_{-3.5}$	$74.4^{+2.6}_{-2.5}$
CW (MPTA)	$3.2^{+2.1}_{-1.5}$	$7.9^{+1.5}_{-1.2}$	$27.0^{+2.7}_{-2.4}$	$51.4^{+2.5}_{-3.2}$
Disc. (CPTA)	2.6	8.8	12.6	—
Disc. (MPTA)	2.6	8.0	7.8	—

CW criterion $\mathrm{SNR} > 2$, discreteness criterion $\ln \mathcal{B} > 2$. For each posterior sample of (N_c, h_c^2) we draw the corresponding single-source strains, evaluate the SNR in each bin, and ask whether the loudest binary clears the threshold.



Goncharov, Sato-Polito, Bi, Zaldarriaga, arXiv:2606.18241.

Why realisation scatter dominates



CPTA, two realisations with *identical* injected $\log_{10} N_c = -1$ and $\log_{10} h_c^2 = -29.3$. Grey inset contours: the noise-free limit.

Left: the posterior collapses to low N_c and excludes the Gaussian bound but do not have enough $\ln \mathcal{B}$ to reject. Right: the N_c marginal disfavors $\log_{10} N_c = 3$. Middle: 2 injected signal compared with the refered power law.



Conclusions

- We built a population-informed prior $p(\rho_k \mid N_c, h_c^2)$ from the compound-Poisson superposition of an SMBHB luminosity function, and ran it through full hierarchical inference on realistic CPTA and MPTA simulations.
- With $N_c = 10$ at yr^{-1}, the CW detection probability ($\mathrm{SNR} > 2$) grows from 4.1% to 45.2% (CPTA) and from 3.2% to 27.0% (MPTA) as the span goes from 5 to 20 yr; discreteness reaches 12.6% (CPTA) and 7.8% (MPTA) at 20 yr.
- The probability of resolving an individual binary at the same significance is higher by a factor of a few if $\log_{10} N_c = 1$: continuous waves come first.
- High-precision data make the posterior geometry *worse* (Neal's funnel at the Gaussian bound); the HMC Slice-within-Gibbs sampler is what makes this campaign feasible and unbiased.
- **Ongoing:** PPTA and EPTA real-data analysis.



Next steps

1. Predict a detection of non-Gaussianity.
2. Predict a detection of CW.
3. Synergy with Holodeck.
4. Incorporate eccentricity and maybe spin effects.
5. Develop a detection ladder.
6. Bayesian model selection.
7. Test detection probability.
8. Is anisotropy required when we marginalize over the brightest source?
9. Transdimensional.
10. Package development of IPTA.
11. Merge your method?

Get in touch with us!

Thanks for your attention!



Introduction

Methodology

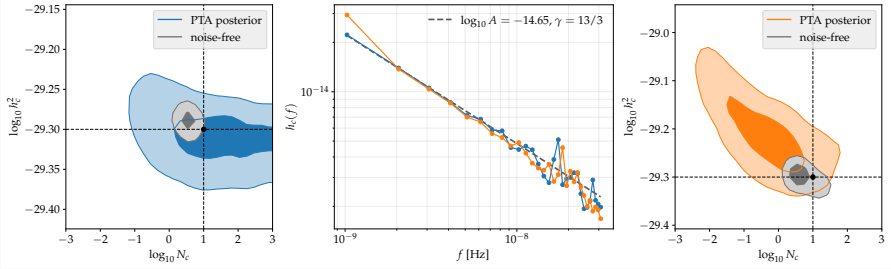
Simulations

Results

*extra



MPTA: the same comparison



MPTA, two realisations with identical injected $\log_{10} N_c = -1$ and $\log_{10} h_c^2 = -29.3$. Grey inset contours: the noise-free limit.

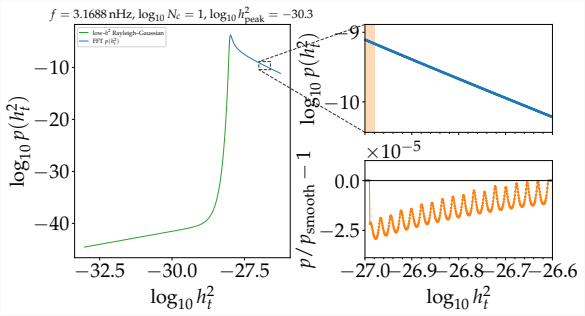
Challenge 1: computational cost

Thousands of hierarchical posteriors are needed: 2 arrays $\times$ 4 spans $\times$ 500 realisations. Every MCMC step must also evaluate the astrophysical prior across all frequency bins.

- PTMCMCSampler: ~ 2 weeks per task on 8 CPUs.
- Discovery with HMC-within-Slice: ~ 4 hours on a single RTX 2070S.
- Gradients from JAX autodiff; NUTS adapts the step size and the mass matrix (curvature) during warm-up.
- $\sim 100\times$ the effective sample size for the NUTS-updated parameters.



Challenge 2: the prior is continuous but not smooth



The compound-Poisson PDF comes from a DFT on finitely many nodes, so $p(h_t^2)$ inherits grid-scale artifacts (bottom right: the residual after smoothing, at the 10^{-5} level).

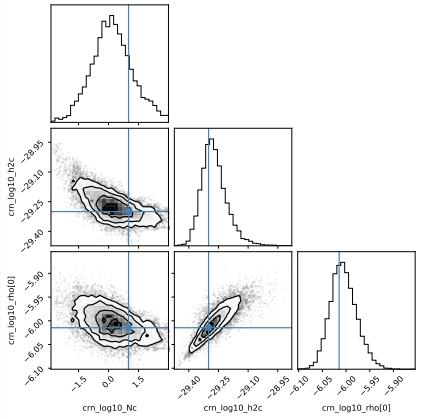
- Tiny in density, but they imprint high-frequency structure on $\nabla \log p$ once summed over 30 bins.
- NUTS then gets trapped in local modes, and warm-up drives the leapfrog step ϵ towards zero.
- Tails are attached analytically so the prior has no hard cutoff:

$$p(h_t^2) \propto (h_t^2)^{-\gamma} \qquad \text{(bright end)},$$

$$p(h_t^2) \propto h_t^2 e^{-h_t^4/2\sigma^2} \qquad \text{(faint end)}.$$



Challenge 3: Neal's funnel at the Gaussian bound



CPTA, 20 yr simulation. As $\log_{10} N_c$ grows, the conditional widths of $\log_{10} h_c^2$ and $\log_{10} \rho_0$ are compressed into a narrow neck at the Gaussian bound.

Why this matters here specifically: the Savage–Dickey ratio is read off *exactly* at that bound. A sampler that stalls at the neck, or never enters it, gets the boundary density wrong and biases the whole forecast.

Fixes.

- Logistic map for bounded coordinates, $\theta(y) = \theta^{\min} + (\theta^{\max} - \theta^{\min})\,\mathrm{sigmoid}(y)$ with $y \sim \mathrm{Logistic}(0, 1)$, so the Jacobian is absorbed.
- Non-centred parameterisation via the probability-integral transform, $\rho_k = Q_k(u_k; \mathbf{\Lambda}) = F_k^{-1}(u_k \mid \mathbf{\Lambda})$ with $u_k \sim \mathrm{U}(0, 1)$: the auxiliary prior no longer depends on $\mathbf{\Lambda}$.

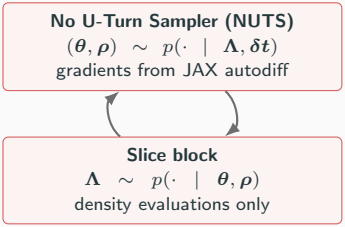


Our sampler: HMC and Slice-within-Gibbs

Each outer **Gibbs** iteration alternates two blocks:

1. **No U-Turn Sampler** (NUTS) for $(\boldsymbol{\theta}, \boldsymbol{\rho})$.
2. **Coordinate-wise slice** sampling for $\boldsymbol{\Lambda} = (N_c, h_c^2)$.

Block alternation decouples the high-dimensional PTA block from the two hyperparameters and avoids moving along the funnel neck jointly.





No U-Turn Sampler (NUTS)

Hamiltonian Monte Carlo augments $\pi(\boldsymbol{\theta})$ with auxiliary momenta and integrates Hamiltonian dynamics with leapfrog steps of size ϵ.

The **No U-Turn Sampler** (Hoffman & Gelman 2011) adaptively chooses the trajectory length: it doubles the leapfrog path until a U-turn criterion signals that further integration would retrace earlier states, then samples a point along the accumulated path.

During warm-up, NUTS jointly adapts ϵ and the mass matrix M from posterior correlations. Gradients come from JAX autodiff.

In our pipeline. With $\boldsymbol{\Lambda} = (N_c, h_c^2)$ held fixed, NUTS updates the PTA parameters $(\boldsymbol{\theta}, \boldsymbol{\rho})$ and never has to move along the hardest $\boldsymbol{\Lambda}$–$\boldsymbol{\rho}$ funnel directions.

Caveat. Because NUTS follows $\nabla \log \pi$, high-frequency roughness in the compound-Poisson prior (Challenge 2) can trap it in local modes and drive $\epsilon \to 0$.

Alternative tested: a *smoothed* reference gradient often gives a larger ESS for (N_c, h_c^2), but its reliability depends on the reference–true gradient mismatch, so it is not our default.



Slice sampling and Gibbs updates

Gibbs sampling updates blocks one at a time from their full conditionals. When a conditional is not available in closed form, any π-invariant kernel—including slice sampling—may replace the exact Gibbs draw.

Slice sampling is gradient-free: given a current state x_0, draw a vertical level $\log y = \log f(x_0) + \log U$ and sample uniformly from $\{x : f(x) \geq y\}$ by shrinking an initial bracket around x_0.

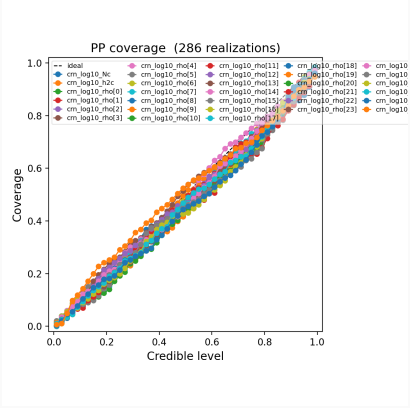
In our pipeline. With $(\boldsymbol{\theta}, \boldsymbol{\rho})$ held fixed, coordinate-wise slice sampling updates $\boldsymbol{\Lambda} = (N_c, h_c^2)$. Only prior evaluations are needed in two dimensions, so the kernel is immune to the FFT-induced non-smoothness.

Settings.

- Bracket width $w = 3$ in raw space.
- Up to 32 shrinkage steps per coordinate (fixed count keeps the kernel JAX-compatible).
- One coordinate-wise sweep over $\boldsymbol{\Lambda}$ per Gibbs iteration.

Together with the logistic and non-centred reparameterisations (Challenge 3), this **HMC&Slice-within-Gibbs** scheme is what makes the hierarchical forecast campaign feasible: ~ 4 h per realisation on a single RTX 2070S versus ~ 2 weeks with PTMCMCSampler on 8 CPUs.

Validation: injection–recovery coverage



P–P coverage over 286 prior-draw realisations, for $\log_{10} N_c$, $\log_{10} h_c^2$ and the 24 free-spectrum amplitudes $\log_{10} \rho_k$.

Thank you

